

# CARLOS TOMÁS MAUTE

## Software Engineer | DevOps

Maputo, Mozambique • +258 86 510 5545 • carlxyzsmaute@gmail.com • linkedin.com/in/maute007 • github.com/Maute007 • maute007.github.io/carlos-maute

### PROFESSIONAL SUMMARY

Software Engineer and DevOps practitioner with 4+ years of experience (since 2022) building and operating mission-critical platforms for the public sector, fintech and e-commerce. I currently develop the Corporate Suite running in production at Mozambique's Ministry of Transport and Logistics (MTL) and its subordinate institutions: an 8-module platform built around a central integration microservice, on infrastructure running Docker, Docker Swarm and Kubernetes. In fintech, I built an in-house KYC module (OCR and computer vision) that replaced an external identity-verification vendor. I led a team of 6 developers delivering platforms for clients in Australia and Kenya, growing to 20,000+ active users in six months. I build with a security mindset (penetration testing, OWASP Top 10, authentication hardening) and take end-to-end ownership: architecture, code, deployment and monitoring.

### TECHNICAL SKILLS

**Languages:** Python, JavaScript, TypeScript, Java, Go (Golang), SQL

**Back-end & APIs:** Django, NestJS, Node.js, Express.js, GraphQL, REST APIs, microservice architecture, systems integration, WebRTC, Java

**Front-end & Mobile:** React, Next.js, Vue.js, Quasar, Bootstrap, HTML5, CSS3, React Native, Flutter

**DevOps & Cloud:** Docker, Docker Swarm, Kubernetes, GitHub Actions (CI/CD), Nginx, Git, Linux server administration, AWS (EC2, S3, Route 53), production monitoring, backups

**Databases:** PostgreSQL, MySQL, SQL Server, MongoDB, Redis, Supabase, Firebase

**Security & Quality:** authorized web-application security testing (Burp Suite, OWASP Top 10), network analysis (Nmap, Wireshark), wireless security (aircrack-ng), authentication hardening (LDAP, MFA, SSO), SonarQube, functional and regression testing

**AI & Automation:** OpenCV, OCR (Pytesseract), AI-powered transcription, n8n (self-hosted), AI-assisted development

**Telecommunications:** Asterisk PBX, VoIP

### PROFESSIONAL EXPERIENCE

#### Software Developer / DevOps

Aug 2025 – Present

Partner Link, Lda. • Maputo, Mozambique • Full-time

- Develop and evolve the **Corporate Suite**, a platform in production at the **Ministry of Transport and Logistics (MTL)** and its subordinate institutions (e.g. FTC), with 8 modules: tasks, meetings, complaints, projects, processes, financial management, attendance and transport.
- Contributed to the architecture of the **central integration microservice**: new systems plug into the suite, which eliminates point-to-point integrations and speeds up go-live.
- Built the tasks and meetings module with real-time **WebRTC** audio and video and **AI-powered automatic transcription of meeting minutes** (native models).
- Run the DevOps layer: containerization with **Docker**, orchestration with **Docker Swarm and Kubernetes**, **CI/CD pipelines in GitHub Actions**, **Nginx** as reverse proxy, Linux server management and production monitoring.
- Implemented the institution's internal phone system with **Asterisk PBX** and a Node.js admin panel, replacing manual call handling and routing.
- Use AI-assisted development (multi-agent orchestration) for prototyping, refactoring and testing, which shortens delivery cycles while keeping human review in place.

Stack: Python/Django, Node.js, React, Bootstrap, PostgreSQL, Docker, Docker Swarm, Kubernetes, Asterisk

#### Software Engineer (Freelance)

2026

Vila Globo • Smart Algarve (fintech) • Portugal • Remote

- Built management platforms for tourism and the municipal sector for clients in Portugal.
- Tourism:** booking and payments system plus a full tourist guide for visitor support.
- Municipal citizen portal:** per-municipality project tracking and public voting, bringing local government closer to citizens.

Backend Developer & QA

2025 (8 months)

Tribox — Paytesy (digital payments fintech) • Maputo, Mozambique • Hybrid

- Built the **KYC module** that validates documents during user registration, using **OCR (Pytesseract)** and **computer vision (OpenCV)** to extract and validate identity-document data.
- The module replaced the external vendor **Sumsu**: the company gained control over the verification process and data, and stopped paying per-verification credits.
- Integrated KYC with the existing financial system through a **Django GraphQL API**.
- Optimized the financial transaction engine (efficiency, reliability and performance under load) and implemented automated payment scheduling, eliminating manual work.
- As QA, ran functional, regression and security tests and maintained static code analysis with **SonarQube**, ensuring stability and regulatory compliance.

Stack: Python/Django, GraphQL, MySQL, OpenCV, Pytesseract, SonarQube

Infrastructure & Server Consultant

2023 – Present

Horizon Consulting / CPTEC • Maputo, Mozambique • Contract

- Manage production Linux servers: monitoring, backups, security hardening and optimization of critical systems, with continuous availability.

Software Development Consultant

2024

AtomTechnology • Mozambique

- Built a tender-management system with Django automation on SQL Server and MySQL, cutting application processing time by **40%**.

Technical Lead — Fullstack Developer

Dec 2022 – 2024

Corner Dev • Maputo, Mozambique • Clients in Australia and Kenya

- Led a **team of 6 developers** in end-to-end development of scalable platforms, growing to **20,000+ active users in six months**.
- **ShopKing** (multi-vendor e-commerce, Australia; MERN stack with Next.js): **35% faster** load times with Redis caching and an automated commission system with **+12% revenue** impact.
- **6Valley** (multi-market e-commerce, Australia): technical consulting and advanced feature development with Next.js, Express.js and MongoDB.
- **Naturali** (marketplace, Kenya): platform for **12,000+ vendors**, with geolocation applied to sustainable deliveries.
- **Linkrider** (corporate mobility app): mobile development in **React Native** with a **NestJS** back-end.
- Industrial production-reporting system in **NestJS; Java** development on startup projects in Kenya.
- Automated business workflows with **self-hosted n8n** and ran **authorized security tests** on the applications (Burp Suite, OWASP Top 10).

Stack: Next.js, React, React Native, NestJS, Express.js, Node.js, Java, MongoDB, Redis, n8n

Fullstack Developer

2023

Eduardo Mondlane University — Innovation Hub • Maputo, Mozambique

- Integrated Own-Revenue Management System: **40%** reduction in tax-processing errors.
- Integrated with **BIM and BCI** banks via RESTful APIs; the system now processes **80% of the university's revenue**.

Web Development Instructor

2023

"Code Like a Girl" Program — Vodacom Mozambique

- Trained **50+ women** in web development and interface design, focused on entering the tech job market.

EDUCATION

BSc in Computer Engineering (in progress) — Technical University of Mozambique (UDM)

2025 – Present

Technical Diploma in IT & Web Programming — Instituto Médio e Politécnico (IMEP)

2020 – 2023

LANGUAGES

Portuguese (native) • English (professional working proficiency) • Changana (fluent)

REFERENCES

Available upon request.